# Directional Coupled Peak Penalties with Original Interval Rows

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**Abstract**

Directional group peaks price the largest weighted departure across selected interval constraints. We derive the box slack-elimination and diagonal-metric proximal identities for coordinate-contraction-monotone penalties, specialize them to directional linear–quadratic group peaks, and describe a native implementation that retains the original interval-row system. Directional row splitting and compact epigraphs already express the same objective; weighted threshold operations and relaxed splitting also have established precedents. The empirical question is whether preserving the original rows reduces complete repeated-forecast computation time against these alternatives. A prospectively registered study retains 210 synthetic instances and 7770 observations across development, heldout and robustness splits. The primary population comprises all 144 heldout instances in six seed clusters, with 32 forecasts and three measured repetitions per method and instance. The median paired reduction score is -212.449875%, with a 95% seed-cluster interval of $[-225.336618\%, -192.873759\%]$, and 9 failed pairs retained under the prescribed scoring rule. The registered 10% improvement is not established. All returned candidate frames satisfy their applicable original-space numerical criteria; iteration limits, unsuccessful common oracles and timeouts remain part of the result. The findings concern synthetic problems on one nonreserved host and do not establish a general speed advantage, operational safety or historical algorithmic priority.

# Chapter 1

# Problem, formulation and related work

## 1.1 Directional interval preferences

Soft constraints represent priced departures from operational preferences. They do not make immutable physical limits optional. A shared group peak couples the departures of several rows: increasing the largest departure can change how a solver allocates corrections to every row in that group. Unequal upper and lower scales express different consequences for the two directions.

Let $P \in \mathbb{R}^{n\times n}$ be symmetric positive semidefinite, $q \in \mathbb{R}^n$ finite, and $A \in \mathbb{R}^{m\times n}$ sparse. Lower and upper interval endpoints may be absent. Disjoint index sets $I_g$ specify soft groups; the remaining rows form the hard set $\mathcal{H}$. Group prices satisfy $a_g, b_g \geq 0$, and the finite directional scales $c_{U,i}, c_{L,i}$ are strictly positive. With $t_+ = \max(t, 0)$, define

$$M_g(z) = \max\left\{0,\ \max_{i\in I_g} c_{U,i}(z_i - u_i)_+,\ \max_{i\in I_g} c_{L,i}(\ell_i - z_i)_+\right\}. \tag{1.1}$$

An absent side contributes no departure. The optimization problem is

$$\begin{aligned} \underset{x\in\mathbb{R}^n}{\text{minimize}} \quad & \tfrac{1}{2}x^\top Px + q^\top x + \sum_g \left(a_g M_g(Ax) + \tfrac{1}{2}b_g M_g(Ax)^2\right), \\ \text{subject to} \quad & \ell_i \leq A_i x \leq u_i \quad (i \in \mathcal{H}). \end{aligned} \tag{1.2}$$

The constraints in $\mathcal{H}$ remain hard. A group with $a_g = b_g = 0$ has zero penalty and is free, rather than hard. Invalid intervals, overlapping or empty groups, nonpositive directional scales and negative curvature are refused by the interface.

## 1.2 Equivalent formulations and attribution

Schwan, Lew and Plancher develop generalized slack projections for soft-constrained quadratic programs without increasing the principal system dimensions [1]. The relation between orthant behavior and coordinatewise monotonicity is established in convex analysis [2]; proximal operations and relaxed splitting are also established tools [3, 4]. Weighted threshold and projection algorithms have substantial prior work [5]. These precedents preclude identifying directional expressibility, a threshold calculation or relaxation alone as a new method.

Directional row splitting replaces an original row by

$$c_{U,i}A_i x \leq c_{U,i}u_i, \qquad -c_{L,i}A_i x \leq -c_{L,i}\ell_i, \tag{1.3}$$

omitting absent sides. The symmetric upper departure peak of these rows equals (1.1). Alternatively, one nonnegative epigraph variable $t_g$ per active group gives the exact compact formulation

$$\begin{aligned} \underset{x,t}{\text{minimize}} \quad & \tfrac{1}{2}x^\top Px + q^\top x + \sum_g (a_g t_g + \tfrac{1}{2} b_g t_g^2), \\ \text{subject to} \quad & t_g \geq 0, \quad c_{U,i}(A_i x - u_i) \leq t_g, \quad c_{L,i}(\ell_i - A_i x) \leq t_g \quad (i \in I_g), \end{aligned} \tag{1.4}$$

together with the original hard rows. Thus the computational comparison must retain both compact epigraphs and directional row splitting. It tests the cost of an original-row implementation, rather than the ability to express the objective.

# Chapter 2

# Slack elimination and the proximal calculation

## 2.1 An exact box identity

Let $C$ be a nonempty closed box, and let $\Psi$ be a proper closed convex nonnegative function with $\Psi(0) = 0$. Write $\Pi_C$ for Euclidean box projection. The contraction property is

$$\Psi(Dw) \le \Psi(w), \qquad D = \operatorname{diag}(d_i), \quad 0 \le d_i \le 1. \tag{2.1}$$

It permits unequal prices on the two rays of a coordinate; it does not require sign symmetry. Define the eliminated interval cost by

$$g_C(z) = \inf_{s \in C} \Psi(z - s). \tag{2.2}$$

**Theorem 2.1** (Box reduction)**.** *Property* (2.1) *is necessary and sufficient for the identity*

$$g_C(z) = \Psi(z - \Pi_C(z)) \tag{2.3}$$

*to hold for every nonempty closed box $C$ and every $z$.*

*Proof.* For any $s \in C$, set $r = z - \Pi_C(z)$. If $z_i > u_i$, then $z_i - s_i \ge r_i > 0$; if $z_i < \ell_i$, then $z_i - s_i \le r_i < 0$. If $z_i$ is inside its interval, $r_i = 0$. Consequently $r = D(z - s)$ for a diagonal contraction $D$. Property (2.1) gives $\Psi(r) \le \Psi(z - s)$, and choosing $s = \Pi_C(z)$ attains equality.

Conversely, fix $w$ and a diagonal contraction $D$. Choose each interval to have endpoints $-w_i$ and $-d_i w_i$, in increasing order, and set $z = 0$. The nearest-box residual is $Dw$, whereas $-w \in C$. Identity (2.3) then implies $\Psi(Dw) \le \Psi(w)$. □

The infimal-convolution representation makes $g_C$ convex, and the continuous residual map in (2.3) gives lower semicontinuity. Corrective slack has the opposite sign to violation residual: in $z + \xi \in C$, its cost is $\Phi(\xi) = \Psi(-\xi)$.

## 2.2 Positive diagonal metrics

For $H = \operatorname{diag}(h_i) \succ 0$, define

$$\operatorname{prox}_f^H(v) = \arg\min_z \left\{ f(z) + \tfrac{1}{2}(z - v)^\top H (z - v) \right\}. \tag{2.4}$$

The metric is fixed during this calculation.

**Proposition 2.2** (Translated residual proximal map)**.** *Under the preceding assumptions,*

$$\operatorname{prox}_{g_C}^{H}(v) = \Pi_C(v) + \operatorname{prox}_{\Psi}^{H}(v - \Pi_C(v)). \tag{2.5}$$

*Proof.* Put $p = \Pi_C(v)$ and clip each candidate coordinate to the segment with endpoints $p_i$ and $v_i$. Diagonality of $H$ makes this clipping nonincreasing for the squared distance from $v$. The clipped point's box residual is a diagonal contraction of the original residual, so its penalty cannot increase either. An optimum may therefore be sought on that product of segments.

On this set, write $z = p + w$ and $r = v - p$. Its box residual equals $w$, so the objective becomes $\Psi(w) + \frac{1}{2}(w-r)^\top H(w-r)$. The unconstrained residual proximal minimizer lies on the same segments between 0 and $r$: coordinate clipping reduces both its distance from $r$ and its penalty. Hence restricting to those segments does not alter that minimum, proving (2.5). □

This is an exact convex-analytic statement. It supplies neither a finite-precision error bound nor an elapsed-time improvement.

## 2.3 Directional peak threshold

For one group, set

$$\Psi(w) = aM(w) + \tfrac{1}{2}bM(w)^2, \qquad M(w) = \max_i \{c_{U,i}(w_i)_+, c_{L,i}(-w_i)_+\}. \tag{2.6}$$

This penalty satisfies (2.1). On the residual segments let $c_i$ be $c_{U,i}$ for $r_i \geq 0$ and $c_{L,i}$ for $r_i < 0$. For a prescribed peak $t \geq 0$, coordinate projection gives

$$w_i(t) = \operatorname{sign}(r_i) \min\{|r_i|, t/c_i\}. \tag{2.7}$$

The remaining scalar convex minimization is

$$\underset{t \geq 0}{\text{minimize}} \quad F(t) = at + \tfrac{1}{2}bt^2 + \tfrac{1}{2}\sum_i h_i(|r_i| - t/c_i)_+^2. \tag{2.8}$$

Sorting the breakpoints $c_i|r_i|$ in descending order identifies an active prefix $S(t) = \{i : c_i|r_i| > t\}$. On an interval with that active set, $F'(t) = 0$ gives

$$t = \frac{\sum_{i \in S(t)} h_i |r_i|/c_i - a}{b + \sum_{i \in S(t)} h_i / c_i^2}. \tag{2.9}$$

The formula is evaluated only with a positive denominator. Empty active sets are handled by the scalar objective and interval endpoints, never by division by zero. The consistent interval and the endpoint $t = 0$ determine the minimizer. If the right derivative at zero is nonnegative, the minimizer is zero. If $a = b = 0$, the proximal map returns the unchanged residual directly. The implementation uses finite local floating-point spacing at threshold boundaries and refuses nonrepresentable arithmetic; an absolute tolerance floor would not establish invariance to units.

## 2.4 Joint conjugates and original-space checks

Define the joint directional dual quantity

$$L_g(y) = \sum_{i \in I_g} \left( \frac{(y_i)_+}{c_{U,i}} + \frac{(-y_i)_+}{c_{L,i}} \right). \tag{2.10}$$

Maximizing $y^\top w$ at fixed peak $M(w) \le t$ gives $tL_g(y)$. Taking the scalar conjugate therefore yields

$$\Psi_g^*(y) = \begin{cases} \dfrac{(L_g(y) - a_g)_+^2}{2b_g}, & b_g > 0, \\ 0, & b_g = 0,\ L_g(y) \le a_g, \\ +\infty, & b_g = 0,\ L_g(y) > a_g. \end{cases} \tag{2.11}$$

This is a group budget, rather than independent coordinate caps. The box support function $\sigma_C$ is also required:

$$g_C^*(y) = \sigma_C(y) + \Psi^*(y), \qquad Px + q + A^\top y = 0. \tag{2.12}$$

A positive multiplier requires a finite upper side and a negative multiplier a finite lower side. Numerical acceptance checks raw and repaired stationarity, bounded dual repair, each group's absolute normalized Fenchel gap and each hard row's complementarity separately. A negative approximate hard-row gap cannot cancel an incorrect soft subgradient. The signed aggregate gap remains diagnostic.

# Chapter 3

# Native splitting, scaling and comparison formulations

## 3.1 Fixed-metric relaxed splitting

Use consensus variables $u = (x, z)$ and $v$, and define

$$F(x, z) = \tfrac{1}{2} x^\top P x + q^\top x + \iota_{\{z=Ax\}}(x, z), \tag{3.1}$$

with $G(v)$ containing the hard intervals and soft costs. The fixed block metric is $\mathcal{M} = \operatorname{diag}(S, H)$, where $S \succ 0$ and $H \succ 0$. In particular,

$$P + S + A^\top H A \succ 0. \tag{3.2}$$

With scaled dual $d$, the four relaxed updates are

$$\begin{aligned} u^{k+1} &= \operatorname{prox}_F^{\mathcal{M}}(v^k - d^k), \\ h^{k+1} &= \alpha u^{k+1} + (1-\alpha) v^k, \\ v^{k+1} &= \operatorname{prox}_G^{\mathcal{M}}(h^{k+1} + d^k), \\ d^{k+1} &= d^k + h^{k+1} - v^{k+1}. \end{aligned} \tag{3.3}$$

Let $R_F = 2\operatorname{prox}_F^{\mathcal{M}} - I$, $R_G = 2\operatorname{prox}_G^{\mathcal{M}} - I$ and $s = v + d$. After a complete step $v = \operatorname{prox}_G^{\mathcal{M}}(s)$, so substitution gives

$$s^{k+1} = (1 - \alpha/2) s^k + (\alpha/2) R_F R_G(s^k), \qquad 0 < \alpha < 2. \tag{3.4}$$

Whitening the fixed metric reduces this to the usual reflected-proximal setting. For proper closed convex $F$ and $G$, exact proximal subproblems and a finite saddle point, the reflected composition is nonexpansive and the strict interior relaxation is averaged. The standard fixed-point and shadow arguments give convergence [3, 4]. The endpoints zero and two are excluded; the penalty prices acquire no $\alpha$ factor. No new convergence-rate theorem is asserted.

The public native implementation fixes a finite scalar $\alpha \in (0, 2)$, defaults to 1.6 and uses direct linear solves. It rejects adaptive penalty updates, automatic OSQP scaling, polishing, directional sensitivities and directional code generation. Its supported implementation scope is narrower than the diagonal-metric identity.

## 3.2 Normalization and workspace invariants

For positive diagonal $D$, consistent row normalization is

$$\bar{A} = DA, \qquad \bar{\ell} = D\ell, \qquad \bar{u} = Du, \qquad \bar{c} = c/D, \qquad (3.5)$$
$$y = D\bar{y}, \qquad H_{\text{physical}} = D\bar{H}D.$$

The bounds must scale with the rows. Setup selects $D_i = \max(c_{L,i}, c_{U,i})$ on soft rows and $D_i = 1$ on hard rows. This selects the corresponding physical metric; it does not preserve the former unscaled-penalty iteration trajectory. The metric, relaxation, $P$, $A$, group membership, prices, directional scales, finite-side pattern and hard-row penalty classification remain fixed within a workspace. The forecast sequence updates $q$ and interval bounds.

Numerical failures clear purported solutions and diagnostics. A quadratic group blocks finite-side violating recession directions. The unboundedness implementation conservatively requires exactly zero computed $Pd$; ambiguous directions remain unresolved. Ambiguous coupled semidefinite blocks larger than the exact 32-variable certification limit are unsupported rather than silently regularized.

## 3.3 Nine registered profiles

Table 3.1 lists the complete comparison set. All five OSQP-style profiles use $\alpha = 1.6$, absolute and relative tolerances $10^{-8}$, at most 200000 iterations and no polishing. Fixed profiles additionally set $\rho = 0.1$, $\sigma = 10^{-6}$, no adaptive penalty and no automatic scaling. The native profile uses its explicit row normalization. Default epigraph and row-split profiles use adaptive penalty and ten scaling iterations. DAQP profiles use absolute tolerance $10^{-8}$ and a 200000-iteration limit; proximal profiles use $\eta_{\text{prox}} = 10^{-10}$. Direct DAQP requires a positive-definite augmented epigraph Hessian. The pinned row-split generalized projection supports the native linear infinity penalty, so squared peaks are declared incompatible. Proximal DAQP preserves the unchanged semidefinite or linear objective. In DAQP 0.10.3, one-shot zero-Hessian and persistent Model policies are distinct. Flags 4 and 5 supply candidates only; independent checks must still accept them [6]. Unsupported objectives remain explicit cells.

Table 3.1: The registered methods and their distinct policies.

| Profile | Formulation and controls |
|---|---|
| `native` | Original-row directional group projection; fixed metric. |
| `epigraph-default` | Stock OSQP compact epigraph; default adaptation/scaling. |
| `epigraph-fixed` | Stock OSQP compact epigraph; fixed controls. |
| `row-split-default` | Directional row-split generalized slack projection; default controls. |
| `row-split-fixed` | Directional row-split generalized slack projection; fixed controls. |
| `daqp-direct` | Direct non-proximal compact epigraph. |
| `daqp-prox-auto` | Proximal epigraph; $\epsilon_{\text{prox}} = -10^{-6}$; one-shot LP policy. |
| `daqp-prox-forced` | Proximal epigraph; $\epsilon_{\text{prox}} = 1$; one-shot LP policy. |
| `daqp-lp-model` | Proximal epigraph; $\epsilon_{\text{prox}} = -10^{-6}$; persistent LP model policy. |

# Chapter 4

# Experimental design and numerical acceptance

## 4.1 Synthetic inputs and complete populations

The study comprises 48 development instances, 144 heldout instances and 18 robustness instances. Each instance supplies 32 forecasts. Every heldout seed has sixteen sparse and eight thermal cells. Sparse cells use $n = 32$ or 96, group size eight, pre-zeroing geometric diagonal spans one or 1000, positive-definite or semidefinite base curvature and $b_g = 0$ or 1. Hard coordinate bounds retain bounded objectives. For semidefinite $P$, the full-matrix condition is infinite; the span label describes the diagonal before zeroing and is not a finite Hessian condition number.

Thermal cells have two independent zones, horizons of twelve or twenty-four one-hour intervals, explicit temperatures and heater-energy inputs. Initial conditions, dynamics and heater limits remain hard. One directional comfort peak is shared over each zone's horizon. The conventional horizon organization follows the standard quadratic-control formulation [7]. For each zone the dynamics have the form

$$T_{k+1} = aT_k + bQ_k + (1 - a)T_{\text{outside},k}. \tag{4.1}$$

Temperature is in degrees Celsius, $Q_k$ is heater energy in kWh per interval, and the objective uses synthetic currency. Retention, gains, weather and prices are authored synthetic values. These are repeated forecasts, not building measurements or independently evolving closed-loop policies.

Robustness separately includes zero-Hessian base cases and a $10^8$ diagonal span, with eighteen instances over three seeds. A zero base Hessian with $b_g > 0$ still produces a quadratic epigraph objective and is not a pure linear program. Development and robustness are descriptive and cannot replace the heldout endpoint.

## 4.2 Measured operation and process supervision

One timed observation includes owned validation, formulation construction and factorization, every forecast update and solve, physical recovery and independent checks. Canonical generation, process/library startup and serialization are excluded symmetrically. Competing compiled libraries are preloaded before the inner timer. Each observation has a fresh supervised process, with workspace reuse only across its own 32 forecasts. One warmup and three measured repetitions are retained per method and instance.

The 30-second worker wall limit includes startup and serialization. Split supervision has a four-hour wall limit and a sampled 4-GiB current/high-water resident-memory limit. These

limits do not reserve CPU or memory. Unrelated host work remains present. The source, dependency map and public SDK/native-wheel identity are registered before execution. No failed case is retuned, reseeded, retried or replaced.

## 4.3 Accuracy in original coordinates

Candidate frames require hard-row error at most $10^{-6}$ in the declared physical units; raw and repaired normalized stationarity, maximum local absolute normalized Fenchel gap, joint dual-budget error and normalized common-oracle objective discrepancy at most $10^{-6}$; and normalized dual repair at most $10^{-7}$. Every forecast must pass. The common stock-OSQP compact-epigraph oracle uses absolute and relative tolerances $10^{-10}$, 200000 iterations and an external $10^{-7}$ criterion.

For example, stationarity uses the worse of the raw and repaired multiplier values of

$$\frac{\left\|Px + q + A^\top y\right\|_\infty}{\max\{1, \|Px\|_\infty, \|q\|_\infty, \|A^\top y\|_\infty\}}. \tag{4.2}$$

Objective discrepancy is normalized by $\max(1, |f_{\text{oracle}}|)$. A group's Fenchel denominator is $\max(1, |\Psi_g|, |\sigma_{C_g}(y_g)|, |y_g^\top A_g x|)$; hard-row denominators are $\max(1, |y_i\,\text{bound}_i|, |y_i(Ax)_i|)$. Dual repair uses $\max(1, \|y\|_\infty)$, and linear joint-budget excess uses $\max(1, L_g, a_g)$. Nonfinite products are rejected before normalization. Bounded repair does not retroactively make the original multiplier an exact member of the dual domain.

## 4.4 Primary inference

For instance $j$ and profile $m$, let

$$T_{j,m} = \text{median}_{r=1,2,3}\, T_{j,m,r}. \tag{4.3}$$

If every compatible measured observation succeeds, define the paired score

$$R_j = 1 - \frac{T_{j,\text{native}}}{\min_{m \neq \text{native},\ m\ \text{compatible}} T_{j,m}}, \tag{4.4}$$

with lexical baseline tie-breaking. Any missing or unsuccessful compatible measured observation instead assigns $R_j = -1$ to the pair and prevents qualification. All 144 instances remain in the population; there is no successful-only denominator.

The bootstrap resamples six whole seeds with replacement, pools their complete 24-instance vectors and takes the median. It uses 10000 resamples and NumPy's linear 2.5th/97.5th percentile limits. The registered prediction requires a median reduction of at least 10%, a strictly positive lower confidence limit and zero failed pairs. The interval concerns the registered seed clusters and observed workload, not universal hardware performance.

# Chapter 5

# Results and retained failures

## 5.1 Primary outcome

The median paired score is -212.449875%, with a 95% seed-cluster interval from −225.336618% to −192.873759% and 9 failed pairs out of 144. The registered prediction fails all three conditions: its estimate is below 10%, its lower confidence limit is negative, and unsuccessful pairs remain. The score combines observed relative times with the prescribed failed-pair value; it is not a uniform runtime multiplier for every case.

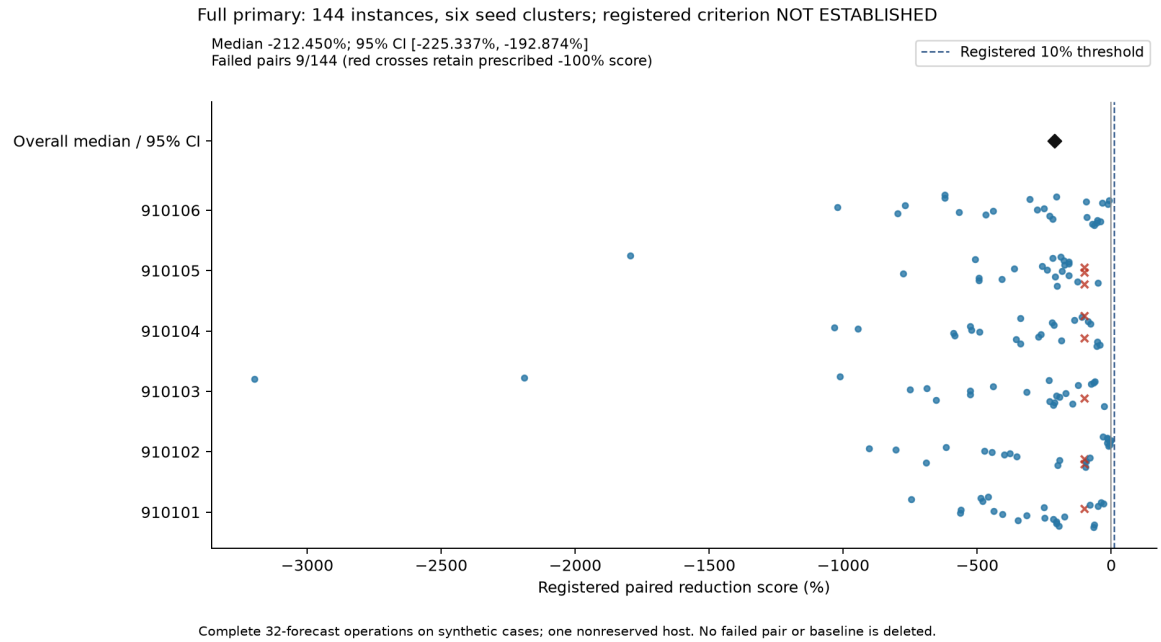



Figure 5.1: All 144 registered paired scores, including the nine unsuccessful pairs at their prescribed −100% score. That value is a failure policy, distinct from observed negative runtime reductions. The full plotted range retains extreme values.

## 5.2 Every measured state

The complete raw journals contain 7770 observations, including warmups and common oracles. Table 5.1 separates the 5670 measured method observations. There are 3667 accepted, twelve accuracy-rejected, 1719 incompatible, 270 dependent oracle-failed and two timed-out measured

observations. The twelve accuracy-rejected batches contain backend iteration limits without returned primal or dual vectors; they are not inaccurate completed KKT solutions.

Table 5.1: Measured method observations; no state is removed from the inventory.

| State | Development | Heldout | Robustness |
|---|---|---|---|
| Accepted | 879 | 2587 | 201 |
| Accuracy-rejected | 0 | 6 | 6 |
| Incompatible | 396 | 1188 | 135 |
| Oracle-failed | 21 | 105 | 144 |
| Timeout | 0 | 2 | 0 |

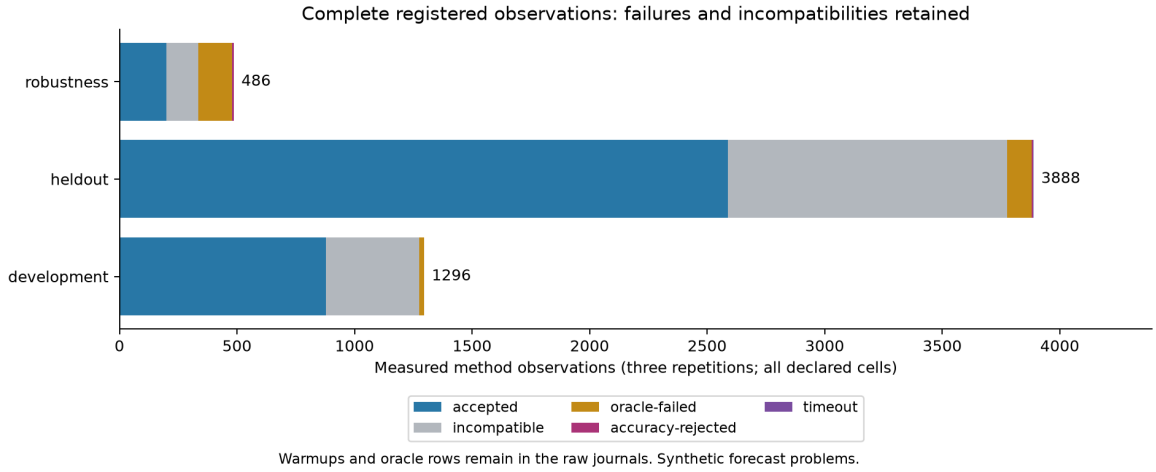



Figure 5.2: Every measured state in the three splits. Warmups and common-oracle records remain in the full journals and are not added to the measured-state bars.

## 5.3 Failure taxonomy and numerical verification

Development has one unsuccessful common oracle, `development-0014` at frame 13, and 21 dependent measured oracle-failed observations. Independent recomputation checks 39021 returned candidate frames, including thirteen valid prefix frames of that unsuccessful oracle; the automatic accepted-batch consumer checks 39008 frames.

Heldout has five common-oracle iteration limits: `heldout-0030`/frame 0, `0054`/22, `0078`/0, `0106`/3 and `0110`/3, all with the `heldout-` prefix. They produce 105 dependent measured oracle-failed observations. The six accuracy-rejected batches are `row-split-default` on `heldout-0014`, frame 3, and `heldout-0026`, frame 22, in all three measured repetitions. Each reaches status 7 at 200000 iterations without candidate vectors; warmups fail identically. Timeouts occur for `native` on `heldout-0095`, repetition index 1, and `daqp-prox-forced` on `heldout-0097`, repetition index 2. Indices are zero-based. Five oracle cases, two row-split cases and two timeout cases account for nine failed primary pairs.

Independent heldout recomputation checks 115124 returned candidate frames, all passing their respective original-space criteria. This includes 248 valid frames from rejected method batches and 28 valid prefixes of failed oracles in addition to 114848 accepted-batch frames. The two counts have different scopes and must not be added together. A killed worker supplies no completed numerical conclusion.

Robustness has six unsuccessful common oracles: `robustness-0000`/frame 21, `0002`/15, `0006`/0, `0008`/4, `0012`/0 and `0014`/13, all with the `robustness-` prefix. The six measured accuracy-rejected batches are `native` and `epigraph-fixed` on `robustness-0003`, each over all three repetitions. Status 7 occurs at frames 1, 24 and 25 after 200000 iterations, without candidate vectors; warmups behave identically. Independent recomputation checks 9245 returned candidate frames, all passing, versus 8960 accepted-batch frames. The six oracle limits produce 144 dependent measured oracle-failed observations.

Across all splits, 163390 returned candidate frames satisfy their applicable checks; 44 noncandidate backend-limit frames remain separate. An unsuccessful common oracle does not by itself prove a candidate algorithm wrong or slower. It makes the registered pair unsuccessful. No oracle substitution, failure deletion or retry changes the primary outcome.

# Chapter 6

# Interpretation, limitations and conclusions

The mathematical identities explain how directional group peaks can retain original interval rows. Exact epigraphs and directional row splitting already express that same objective. Preserving principal dimensions alone therefore does not establish a computational advantage. The complete-operation measurements refute the registered runtime prediction for the observed workload.

The earlier separable study [8] is a different 456-instance computation. Its median reduction was 0.499839%, with a 95% interval of $[-3.291967\%, 3.944835\%]$, which also failed its 10% prediction. These populations, endpoints and penalty families cannot be pooled into an improvement claim.

The current results are limited to authored synthetic sparse problems and thermal forecasts, nine registered profiles, three measured repetitions and six primary seed clusters on one nonreserved Windows host. They are not field energy savings, closed-loop safety evidence or a universal runtime comparison. Factorization costs and allocations were not isolated causal effects. Conservative numerical refusal, iteration limits and process supervision are part of the reported behavior.

The fixed-metric splitting argument assumes a finite saddle point and does not cover every adaptive implementation. Finite floating-point arithmetic needs separate numerical checks. Larger ambiguous semidefinite cases outside the certification scope remain unsupported. Neither the derivations nor the tests establish global algorithmic novelty. The public engine is an engineering distribution, not a certificate of operational suitability.

The contribution is an inspectable original-row directional implementation, its bounded mathematical formulation and a reproducible negative comparison. The registered 10% speed improvement is not established. Any future computational prediction requires its own justified protocol rather than a changed interpretation of these retained observations.

## Data and code availability

The full synthetic inputs, original-space oracle/method journals, failures, warmups, CSV, protocol and registered source snapshots are available in the supporting dataset [9]. Its ZIP SHA256 is `45b4eceda69f80a30a4ce349d0e9baefacace0de394fe315f3c392260ea8d418`. The scientific source is commit `246bf87e690f4d010494c556e2021d311d587422` of `https://github.com/fsantibanezleal/CAOS_SoftQP`. The independent engine source is `https://github.com/fsantibanezleal/CAOS_SoftQP_Engine`; the study consumes `softqp==0.2.0` from `https://pypi.org/project/softqp/0.2.0/`. No external operational dataset or private records are needed. Source-code licenses remain separate from the text

license. The compiled document's source and build instructions are included in the public repository.

## Declaration of generative AI use

Generative AI tools were used for mathematical development, software implementation, document preparation and independent verification assistance. The author is responsible for reviewing and verifying all content and takes full responsibility for the work.

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